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# Design and Reception Method of Code Shift Keying Signals With Power-of-Two Phase Intervals

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## ABSTRACT

Precise point positioning (PPP) technology has garnered extensive attention for its ability to deliver real-time centimetre-level accuracy services. When providing PPP services via satellite navigation signals, it is necessary to increase the data rate to over 1000 bps. Code shift keying (CSK) technology has emerged as a key candidate for satellite-based PPP signal design. It can increase the data rate without requiring additional frequency resources and compromising ranging performance. However, CSK technology faces several challenges. Firstly, it is vulnerable to multipath interference, which can lead to intersymbol interference. Secondly, the complexity of demodulation increases exponentially with an increase in the number of bits per symbol. In this paper, a novel approach for the design and reception of CSK signals with a phase interval mapping of power-of-two chips is proposed. Analyses demonstrate that this method is capable of significantly reducing the demodulation complexity while mitigating the intersymbol interference. The CSK (6, 2) modulation with a phase interval mapping of 128 chips is employed for the BDS PPP-B2b signal. The signal is demodulated using the FFT with a phase interval output of 128 points. The computation can be reduced by approximately 85%, compared to the continuous phase mapping CSK signal demodulated by partial-output FFT.

## 1 | Introduction

Global navigation satellite systems (GNSS), serving as a pivotal infrastructure for the provision of spatiotemporal reference services, have experienced a surge in their applications across various sectors, including civil aviation, intelligent transportation, geodesy and daily life. The rapid advancement of unmanned aerial vehicles, autonomous driving technologies and smartphones, coupled with the maturation of high-precision positioning technologies, has led to an escalating demand for precise point positioning (PPP) services with decimetre or centimetre accuracy. In response to this demand,

major GNSS systems have initiated the integration of satellite-based centimetre-level PPP services into the development plans for their forthcoming generations [1–8].

PPP services facilitate high-precision positioning for users by broadcasting real-time GNSS correction data, encompassing precise ephemeris and phase biases. To support centimetre-level PPP, a data transmission rate exceeding 1000 bps is imperative. Conventional GNSS signals, employing BPSK-modulated messages, are typically constrained by limited frequency resources and power, resulting in data rates that range from 50 bps to 500 bps. Consequently, attaining data rates above 1000 bps

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necessitates the adoption of novel signal design methodologies [9]. To enhance data rates without the allocation of additional frequency resources or detriment to ranging capabilities, code shift keying (CSK) technology has emerged as a focal point of interest among GNSS signal designers [9–14].

In recent years, numerous researchers have extensively explored the application of CSK technology in GNSS signal design. Studies by Garcia Peña et al. [9, 10] and Jin Shuxin et al. [11] have provided comprehensive insights into the feasibility of CSK for future GNSS signals. R'emi Chauvat et al. [12], Jin Shuxin et al. [13] and Fu Yuehui et al. [14] investigated the integration of CSK with multilevel LDPC coding for navigation signal design. Japan's QZSS system has adopted CSK modulation on its L6D and L6E signals, achieving a data transmission rate of 2000 bps [3].

Higher-order CSK modulation, which increases the number of bits per symbol, enhances transmission efficiency. However, compared to BPSK, it introduces significantly higher demodulation complexity and is more susceptible to intersymbol interference (ISI). For example, 6-bit CSK symbols require simultaneous matching with 64 parallel correlators; increasing to 7 bits necessitates 128 parallel correlators. Consequently, the receiver complexity increases exponentially with the modulation order. Moreover, CSK typically maps data onto a contiguous phase of the PRN code. In the presence of multipath, this mapping results in ISI and increased bit error rates [15].

To address the high complexity in CSK reception, an efficient demodulation algorithm utilising a partial-output fast Fourier transform (FFT) was proposed [16]. By converting time-domain cyclic correlation into frequency-domain multiplication and applying inverse FFT only to the mapped phase positions, the algorithm significantly reduces computation. Because the number of modulated phases is much smaller than the PRN code length, this partial-output FFT approach offers substantial savings in demodulation complexity.

To mitigate ISI, a phase-discontinuous CSK message transmission method was proposed, wherein message bits are mapped using equal phase intervals [15]. This approach helps eliminate multipath interference within the phase mapping region. However, when using partial-output FFT for demodulation of such phase-discontinuous CSK signals, the inverse FFT must compute all correlation values from the first to the last mapped phase, rendering its computation nearly equivalent to that of full FFT, thereby negating its efficiency advantage.

To address the aforementioned limitations, this paper proposes a novel approach for the design and reception of CSK signals with a phase interval of power-of-two chips, building upon phase-discontinuous equal-interval mapping. Theoretical analysis and simulation results demonstrate that the proposed method alleviates intersymbol interference while significantly reducing demodulation complexity—exceeding both traditional FFT and partial-output FFT approaches. The CSK (6, 2) modulation with a phase interval mapping of 128 chips is employed for the BDS PPP-B2b\_Q signal. The signal is demodulated using FFT with a phase interval output of 128 chips. The computation can be reduced by approximately 85% compared to the

continuous phase mapping CSK signal demodulated by partial-output FFT.

## 2 | Fundamentals of CSK Signals

### 2.1 | Code Shift Keying Modulation

Code shift keying is an M-ary orthogonal modulation scheme. Its fundamental principle involves generating  $M(M = 2^K)$  new spreading sequences through cyclic shifts of a single base spreading sequence. These cyclically shifted sequences are used to represent different data symbols, thereby enabling the modulation of K bits of information using a single spreading sequence [17, 18].

CSK-modulation is generally denoted as CSK(K), where K represents the number of information bits modulated by a single CSK symbol. Let  $c_d(n)$  denote the base spreading sequence,  $c_d(t)$  denote the base spreading modulated symbol,  $c_x(n)$  denote the x-th cyclically shifted spreading sequence, and  $c_x(t)$  denote the x-th CSK-modulated symbol. Then, the CSK symbol model can be expressed as follows:

$$c_x(t) = c_d(\text{mod}(t + m_x T_c, N_{PRN} \cdot T_c)) = \sum_{n=0}^{N_{PRN}-1} c_x(n) \cdot \text{rect}\left(\frac{t - nT_c}{T_c}\right) \quad (1)$$

$$c_x(n) = c_d(\text{mod}(n + m_x, N_{PRN})), x = 0, \dots, M - 1, n = 0, \dots, N_{PRN} - 1 \quad (2)$$

where

$N_{PRN}$ : The length of the spreading sequence (PRN code);

$T_c$ : The chip duration of the spreading sequence;

$m_x$ : The phase shift of the x-th cyclically shifted spreading sequence relative to the base spreading sequence;

$M$ : The total number of cyclically shifted spreading sequences,  $M = 2^K$ , with  $M$  being less than  $N_{PRN}$ .

Repeated code shift keying (R-CSK) modulation transmits the same CSK modulation symbol by repeatedly using cyclically shifted spreading sequences. This technique enhances the symbol energy per transmitted unit and reduces the bit error rate (BER) in data transmission [19]. R-CSK modulation can be denoted as CSK(K, N), where K denotes the number of information bits carried by a single CSK symbol and N indicates that one CSK symbol is composed of N repeated spreading sequences. In the R-CSK scheme, each spreading sequence modulates K/N bits of data. Under the same data transmission rate and signal power, R-CSK exhibits better BER performance than conventional CSK [19]. A single CSK(K, N) modulation symbol can be expressed as follows:

$$s_x(t) = \sum_{n=0}^{N-1} c_x(t - nT_{PRN}) \quad (3)$$

where  $s_x(t)$  denotes the  $x$ -th CSK( $K, N$ ) modulation symbol.  $T_{PRN} = N_{PRN} \cdot T_c$ ,  $T_{PRN}$  is the period of the spreading sequence.

## 2.2 | Modulation Information Mapping Method of CSK Signals

Theoretically, it is sufficient for the  $K$ -bit message modulated by the cyclic shift PRN code  $c_x(n)$  and the phase shift value of the PRN code  $m_x$  to meet the one-to-one mapping requirement. In practical applications, to simplify the demodulation process at the receiver, it is common to adopt a method where the phase shift value of the cyclic shift PRN code is equal to the decimal value of the modulated  $K$ -bit message. Specifically, different phase values are assigned to each symbol of the modulation message. In this way, the modulation message is mapped to a continuous phase region of the PRN code (either with an advanced phase order or a lagging phase order) [14, 16]. This is illustrated in Figure 1.

In the context of multipath interference, the overlay of multipath signals with direct signals can disrupt the decision-making process of signal demodulators, thereby inducing ISI. When the delay time of multipath signals is an integer multiple of the chip time, the impact of ISI becomes particularly significant, leading to an elevation in the bit error rate during the CSK demodulation process. To address this challenge, a novel approach for disseminating messages using phase-discontinuous CSK signals was proposed [15]. This method employs a modulation-information mapping strategy with equally spaced phase intervals to modulate message data, as illustrated in Figure 2 (where  $Np$  denotes  $N_{PRN}$  and  $Z$  denotes the phase interval). When the multipath delay time satisfies the condition  $\geq 1$  chip and  $\leq Z-1$  chips, within this delay range, there are no mapped phases corresponding to the modulation message. Consequently, ISI caused by multipath signals can be effectively avoided.

| Information symbol index | PRN code phase |       |     |          |        |        |  |
|--------------------------|----------------|-------|-----|----------|--------|--------|--|
| 0 $\Rightarrow$          | PRN(0)         | 0     | 1   | ...      | $Np-2$ | $Np-1$ |  |
| 1 $\Rightarrow$          | PRN(1)         | 1     | 2   | ...      | $Np-1$ | 0      |  |
| $\vdots$                 | $\vdots$       |       |     | $\vdots$ |        |        |  |
| $M-1 \Rightarrow$        | PRN( $M-1$ )   | $M-1$ | $M$ | ...      | $M-3$  | $M-2$  |  |

(a) Advanced phase order (The information mapping with a phase-increment order for phase-continuous CSK modulation)

| Information symbol index | PRN code phase  |          |          |          |          |        |
|--------------------------|-----------------|----------|----------|----------|----------|--------|
| 0 $\Rightarrow$          | PRN(0)          | 0        | 1        | ...      | $Np-2$   | $Np-1$ |
| 1 $\Rightarrow$          | PRN( $Np-1$ )   | $Np-1$   | 0        | ...      | $Np-3$   | $Np-2$ |
| $\vdots$                 | $\vdots$        |          |          | $\vdots$ |          |        |
| $M-1 \Rightarrow$        | PRN( $Np-M+1$ ) | $Np-M+1$ | $Np-M+2$ | ...      | $Np-M+1$ | $Np-M$ |

(b) Lagging phase order (The information mapping with a phase-decrement order for phase-continuous CSK modulation)

**FIGURE 1** | Schematic of information mapping for phase-continuous CSK modulation ( $Np$  denotes  $N_{PRN}$ ).

## 2.3 | Demodulation of CSK Signals

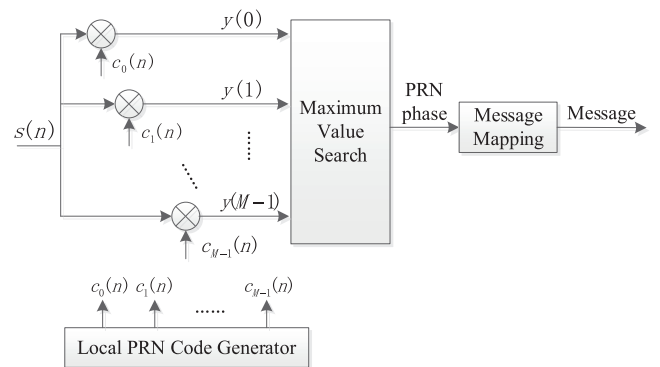
In contrast to BPSK-modulated signals, CSK-modulated signals can effectively increase the data transmission rate. However, this enhancement is accompanied by a significant increase in the complexity of information demodulation at the receiver. There are typically two methods for demodulating CSK signals: time-domain demodulation and frequency-domain demodulation.

The time-domain demodulation method for the CSK signal involves performing cyclic shift correlation calculations on the received baseband signal sequence  $s(n)$  and  $M$  locally generated cyclic shift spreading sequences  $c_x(n)$ , and then outputting the correlation values  $y(x)$ . The modulation message demodulated is the information mapped by the cyclic shift spreading sequence corresponding to the maximum correlation value  $MAX(y(x))$ .  $s(n)$  is a baseband digital signal sequence that has been synchronised and carrier-stripped. Equation (4) presents the calculation formula for time-domain correlation. Figure 3 illustrates a typical time-domain demodulation procedure of CSK signals. Notably, the higher the modulation order of the CSK signal, meaning the greater the number of bits in the spreading modulation symbol, the higher the complexity of signal reception. For example, 6-bit CSK symbols require simultaneous matching with 64 parallel correlators; increasing to 7 bits necessitates 128 parallel correlators. The QZSS L6 signal employs 8-bit modulation, requiring 256 parallel correlators. Consequently, the complexity of signal reception increases exponentially as the number of bits per modulation symbol grows.

$$y(x) = \sum_{m=0}^{N_{PRN}-1} s(m)c_x(m) = \sum_{m=0}^{N_{PRN}-1} s(m)c_d(\text{mod}(m + m_x, N_{PRN})) \quad (4)$$

| Information symbol index | PRN code phase     |             |               |     |               |               |  |
|--------------------------|--------------------|-------------|---------------|-----|---------------|---------------|--|
| 0 $\Rightarrow$          | PRN(0)             | 0           | 1             | ... | $Np-2$        | $Np-1$        |  |
| 1 $\Rightarrow$          | PRN( $Z$ )         | $Z$         | $Z+1$         | ... | $Z-2$         | $Z-1$         |  |
| 2 $\Rightarrow$          | PRN( $2 \cdot Z$ ) | $2 \cdot Z$ | $2 \cdot Z+1$ | ... | $2 \cdot Z-2$ | $2 \cdot Z-1$ |  |
| $\vdots$                 | $\vdots$           | $\vdots$    |               |     |               |               |  |
| $M-1 \Rightarrow$        | PRN( $M \cdot Z$ ) | $M \cdot Z$ | $M \cdot Z+1$ | ... | $M \cdot Z-2$ | $M \cdot Z-1$ |  |

**FIGURE 2** | Schematic of information mapping for phase-discontinuous modulation with equal interval.



**FIGURE 3** | Flowchart of CSK signal time-domain demodulation method.

where  $m_x$  denotes the phase of the cyclic shift,  $x = 0, \dots, M - 1$ ;  $M$  represents the total number of cyclically shifted spreading sequences, and  $N_{PRN}$  refers to the length of the spreading sequence (PRN code).

The frequency-domain demodulation method for CSK signals involves performing fast Fourier transform (FFT) on the synchronised, carrier-stripped baseband signal sequence  $s(n)$  and the basic spreading sequence  $c_d(n)$  separately to obtain the frequency-domain sequences  $S(k)$  and  $C_d(k)$ . After multiplying these sequences in the frequency-domain, the resulting frequency-domain correlation value  $Y(k)$  is obtained. Then, an inverse fast Fourier transform (IFFT) is applied to derive the correlation sequence  $y(n)$ . The message demodulated is the information mapped by the cyclic shift spreading sequence corresponding to the maximum correlation value  $MAX(y(n))$ . Equations (5) and (6) provide the formulae for frequency-domain demodulation, and Figure 4 illustrates a typical frequency-domain demodulation procedure of CSK signals.

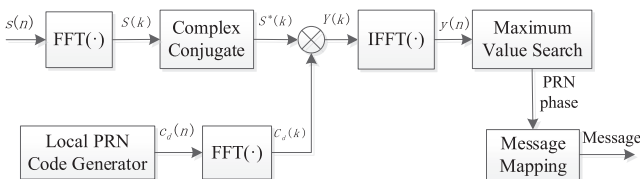
$$Y(k) = S^*(k) \cdot C_d(k) \quad (5)$$

$$y(n) = IFFT[Y(k)] = IFFT[S^*(k) \cdot C_d(k)] \quad (6)$$

In the frequency-domain demodulation method of CSK signals, the inverse fast Fourier transform (IFFT) simultaneously yields the correlation values corresponding to all possible code phase shifts of cyclic shift spreading sequences. However, for a  $K$ -bit CSK modulation, only  $M$  code phase shifts are valid, where  $M = 2^K$ , and typically  $M$  is much smaller than the total number  $N_{PRN}$  of phase positions in the basic spreading sequence  $c_d(n)$ . Therefore, in CSK signal demodulation, it is unnecessary to compute all IFFT outputs. Instead, only the correlation values corresponding to the  $M$  valid code phase positions need to be computed, enabling the use of a partial-output IFFT computation to reduce the overall demodulation complexity. To this end, a partial-output FFT-based CSK demodulation algorithm was proposed specifically for CSK signals employing continuous phase mapping [16]. During the IFFT stage of frequency-domain demodulation, only a continuous set of  $M$  outputs corresponding to valid code phases are computed.

The number of computing nodes for the butterfly operations of the partial-output FFT for CSK signals employing phase-continuous mapping is given in Equation (7) [16].

$$T_1 = K \times 2^L + \sum_{i=1}^{L-K} 2^{L-i} = K \times 2^L + 2^L - 2^K \quad (7)$$



**FIGURE 4** | Flowchart of CSK signal frequency-domain demodulation method.

where  $2^L$  denotes the total number of FFT points,  $L$  represents the number of butterfly operation stages, and  $K$  is the number of information bits per CSK modulation symbol.

The proportion of butterfly computation nodes required by the phase-continuous partial-output FFT relative to that of the conventional full-output FFT is given in Equation (8) [16].

$$\eta_1 = \frac{T_1}{L \times 2^L} = \frac{1}{L} (K + 1 - 2^{K-L}) \quad (8)$$

When the partial-output FFT demodulation algorithm is applied to the demodulation of CSK signals employing phase-discontinuous mapping, the  $2^K$  correlation values of individual code phases are evenly distributed across the entire phase range of the spreading sequence. Consequently, the IFFT must output all correlation values between the first and last mapped phases, leading to a computation comparable to that of conventional full-output FFT demodulation. This significantly reduces the efficiency of the demodulation process.

### 3 | Modulation and Demodulation of CSK Signals With Power-of-Two Phase Intervals

#### 3.1 | Modulation of CSK Signals With Power-of-Two Phase Intervals

The CSK signals with power-of-two phase intervals refers to the CSK signal where the cyclically shifted spreading code is mapped to the modulation data bits with a phase interval of  $Z = 2^P$  chips (where  $P$  is a natural number), as shown in Figure 5 (where  $N_p$  denotes  $N_{PRN}$ ).

The value of  $P$  should be chosen such that the phase interval  $Z$  is as large as possible in order to maximally alleviate the inter-symbol interference caused by multipath signals. At the same time, considering the cyclic shift characteristics of CSK spreading code phases, the phase interval between the last phase output point and the first phase output point should be greater than or equal to  $2^P$  chips, as presented in Equations (9) and (10).

$$N_{PRN} - (M - 1) \cdot 2^P \geq 2^P \quad (9)$$

$$P \leq \log_2(N_{PRN}) - \log_2(M) = \log_2(N_{PRN}) - K \quad (10)$$

Considering that the phase interval should be as large as possible,  $P$  is selected according to Equation (11).

| Information symbol index | PRN code phase       |               |                   |     |                   |                   |
|--------------------------|----------------------|---------------|-------------------|-----|-------------------|-------------------|
| 0                        | PRN(0)               | 0             | 1                 | ... | $N_p - 2$         | $N_p - 1$         |
| 1                        | PRN( $2^P$ )         | $2^P$         | $2^P + 1$         | ... | $2^P - 2$         | $2^P - 1$         |
| 2                        | PRN( $2 \cdot 2^P$ ) | $2 \cdot 2^P$ | $2 \cdot 2^P + 1$ | ... | $2 \cdot 2^P - 2$ | $2 \cdot 2^P - 1$ |
| ...                      | ...                  | ...           | ...               | ... | ...               | ...               |
| $M-1$                    | PRN( $(M-1)2^P$ )    | $(M-1)2^P$    | $(M-1)2^P + 1$    | ... | $(M-1)2^P - 2$    | $(M-1)2^P - 1$    |

**FIGURE 5** | Information mapping of power-of-two phase interval CSK modulation.



$$P = \lfloor \log_2 N_{PRN} \rfloor - K \quad (11)$$

The phase interval and spreading code mapping phases are configured as follows:

1. Let  $P = \lfloor \log_2 N_{PRN} \rfloor - K$ , and the code phase interval  $Z = 2^P$ .
2. The phase indices  $0, 2^P, 2 \cdot 2^P, 3 \cdot 2^P, \dots, (M-2) \cdot 2^P, (M-1) \cdot 2^P$  are taken as the CSK cyclically shifted spreading code mapping phase configuration for  $M$  information symbols.

### 3.2 | Demodulation of CSK Signals With Power-of-Two Phase Intervals

In the demodulation of CSK signals with power-of-two phase intervals, only  $M$  code phase shifts corresponding to the mapped information are possible. Consequently, in the frequency-domain demodulation of CSK signals with power-of-two phase intervals, the inverse fast Fourier transform (IFFT) needs to output only the correlation values of the code phase shifts with power-of-two phase intervals that map the modulated information. Figure 6 illustrates a typical frequency-domain demodulation procedure of CSK signals.

The computational principle of the fast Fourier transform (FFT) is to divide the  $N$ -point ( $N = 2^L$ ) input sequence  $x(n)$  into two subsequences based on the parity of the point indices. The  $N$ -point output sequence  $X(k)$  is obtained by performing a butterfly operation on the discrete Fourier transforms (DFTs) of the two subsequences. This process can be recursively applied to decompose the sequence into 2-point DFTs, thereby forming the butterfly operation structure shown in Figure 7.

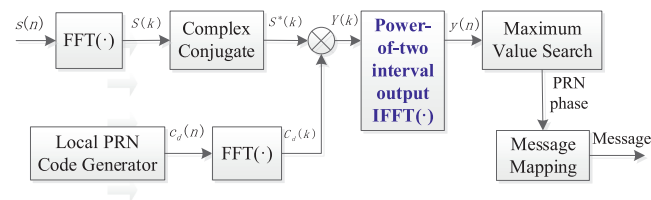
The basic butterfly operation formula is as follows in Equations (12) and (13):

$$X(k) = X_1(k) + W_N^k \cdot X_2(k) \quad (12)$$

$$X\left(k + \frac{N}{2}\right) = X_1(k) - W_N^k \cdot X_2(k) \quad (13)$$

where  $k = 0, 1, 2, \dots, 2^{L-1} - 1$ ,  $X_1(k)$  and  $X_2(k)$  are the DFTs of the even and odd subsequences, respectively, with  $2^{L-1}$  points.

It is observed that for the output phase interval  $2^{L-1}$ , the two points that belong to the same butterfly operation unit are output. With each subsequent level of decomposition, the phase interval between the two points output by the same butterfly



**FIGURE 6** | Flowchart of frequency-domain demodulation method for CSK signals with power-of-two phase intervals.

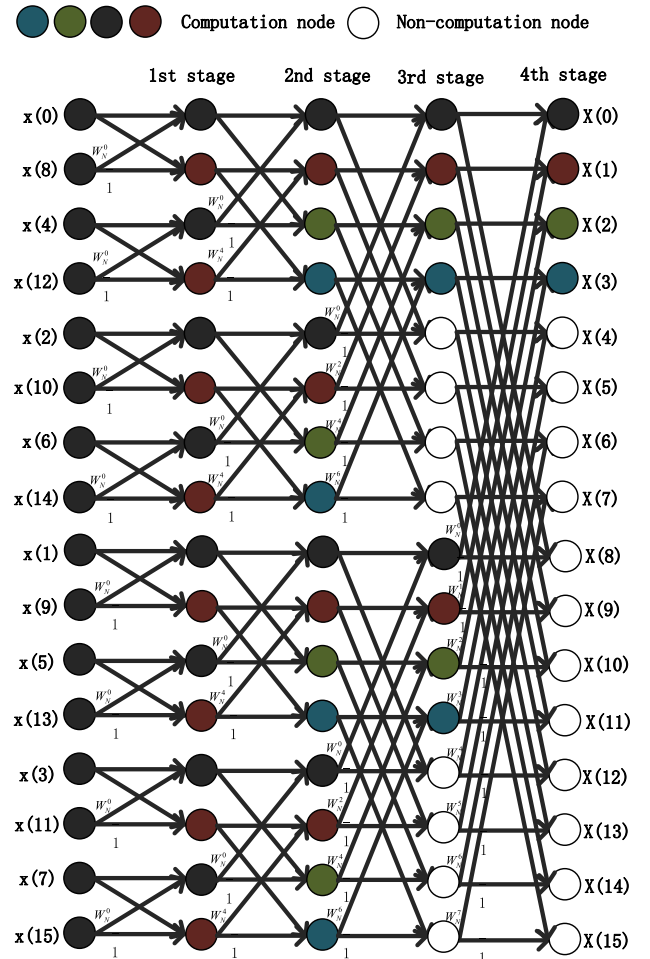
unit decreases by a factor of 2. If the output points are selected at power-of-two phase intervals, the characteristics of the butterfly operation unit can be fully utilised, thereby reducing the complexity of demodulation. According to this idea, this paper proposes a power-of-two phase interval output FFT method for demodulating CSK signals with power-of-two phase intervals. That is, when demodulating CSK signals using a frequency-domain method, only  $M$  interval correlation values corresponding to the  $2^P$  phase interval are computed during IFFT.

The power-of-two phase interval output FFT comprises the following primary steps:

Step 1: To facilitate the application of the FFT algorithm, the synchronised and carrier-stripped baseband signal sequence  $s(n)$  and the base spreading sequence  $c_d(n)$  are first zero-padded to extend their lengths to the nearest power-of-two length  $N$ , where  $N = 2^L$  and  $L = \lceil \log_2 N_{PRN} \rceil$ .

Step 2: Apply FFT to the zero-padded sequences  $s(n)$  and  $C_d(k)$ , transforming them into their respective frequency-domain representations  $S(k)$  and  $C_d(k)$ , respectively.

Step 3: Multiply the conjugate of  $S(k)$  by  $C_d(k)$  in the frequency-domain to obtain the frequency-domain correlation result  $Y(k)$ .



**FIGURE 7** | 16-point FFT operation structure with phase-continuous partial output (4 points output).

Step 4: Perform an IFFT on  $Y(k)$ , using power-of-two phase interval outputs to compute the correlation result  $y(n)$ . The output indices for the correlation values are selected according to the following scheme:

1.  $K = 1$ , output 2 points with a phase interval of  $2^{L-1}$  chips, and compute the 0-th and  $2^{L-1}$ -th points accordingly;
2.  $K = 2$ , output 4 points with a phase interval of  $2^{L-2}$  chips, and compute the 0-th,  $2^{L-2}$ -th,  $2 \cdot 2^{L-2}$ -th,  $3 \cdot 2^{L-2}$ -th points accordingly;

3.  $K = 3$ , output 8 points with a phase interval of  $2^{L-3}$  chips, and compute the 0-th,  $2^{L-3}$ -th,  $2 \cdot 2^{L-3}$ -th,  $3 \cdot 2^{L-3}$ -th,  $4 \cdot 2^{L-3}$ -th,  $5 \cdot 2^{L-3}$ -th,  $6 \cdot 2^{L-3}$ -th,  $7 \cdot 2^{L-3}$ -th points accordingly;
4. Output  $M = 2^K$  points with a phase interval of  $2^{L-K}$  chips, and compute the 0-th,  $2^{L-K}$ -th,  $2 \cdot 2^{L-K}$ -th,  $3 \cdot 2^{L-K}$ -th, ...,  $(2^K - 2) \cdot 2^{L-K}$ -th,  $(2^K - 1) \cdot 2^{L-K}$ -th points accordingly.

## 4 | Demodulation Computation Analysis of CSK Signal Based on FFT

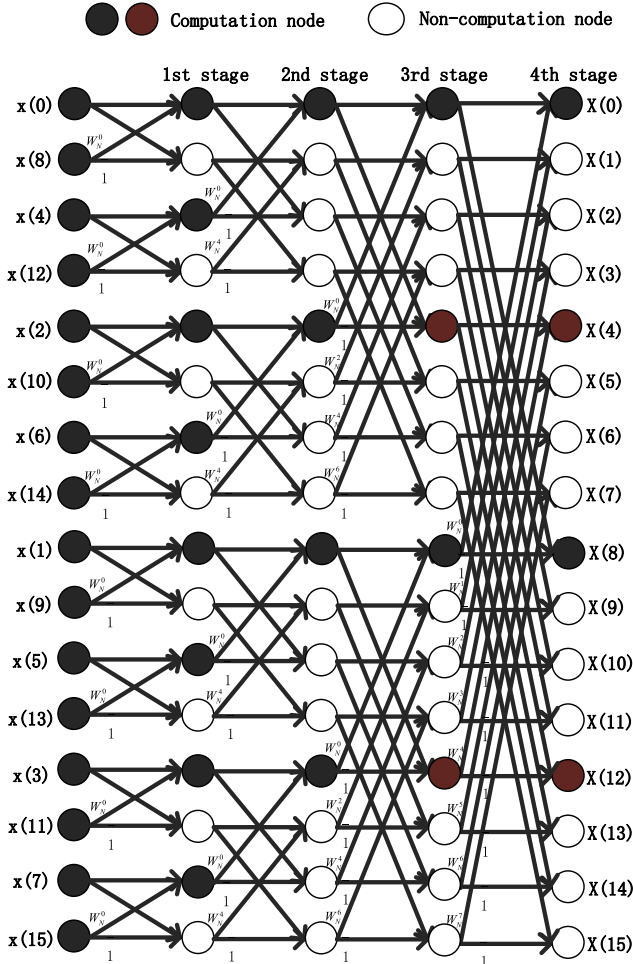
### 4.1 | Computation of 16-Point FFT With Power-of-Two Phase Interval Output

To analyse the computational complexity of the power-of-two phase interval partial-output FFT demodulation, we begin with a 16-point FFT as an example. A 16-point FFT comprises four stages of butterfly operations and 64 computation nodes. Figure 7 illustrates the schematic of the computation nodes for the phase-continuous partial-output FFT with 16-point butterfly operations. Figure 8 illustrates the schematic of the computation nodes for the phase-discontinuous power-of-two phase interval output FFT with 16-point butterfly operations.

Table 1 offers a comparative analysis of the number of computation nodes for different output points of the continuous phase partial output and power-of-two phase interval output of 16-point FFT butterfly operations. With a 4-point continuous phase output, 44 computation nodes are required, which accounts for 68.75% of the total nodes. In contrast, with a 4-point power-of-two phase interval output, only 20 computation nodes are needed, constituting 31.25% of the total nodes. This clearly indicates that the demodulation complexity of the power-of-two phase interval partial-output FFT is significantly reduced.

### 4.2 | Computation of Power-of-Two Phase Interval Output FFT

Based on the characteristics of the butterfly operation, the FFT computational structure of the power-of-two phase interval output FFT can be further optimised to reduce the computational complexity. By recursively analysing the computation nodes shown in Figure 7 and Table 1, the distribution pattern of



**FIGURE 8** | 16-point FFT operation structure with phase-discontinuous power-of-two interval output (4 points output).

**TABLE 1** | Comparison of computation nodes between phase-continuous partial-output FFT and power-of-two interval output FFT ( $L = 4$ ,  $2^L = 16$ ).

| $M (M = 2^K)$ |                      | Computation nodes per stage |     |     |     | Total | Proportion |
|---------------|----------------------|-----------------------------|-----|-----|-----|-------|------------|
|               |                      | 1st                         | 2nd | 3rd | 4th |       |            |
| 2             | Phase-continuous     | 16                          | 8   | 4   | 2   | 30    | 46.88%     |
|               | $2^3$ phase interval | 8                           | 4   | 2   | 2   | 16    | 25.00%     |
| 4             | Phase-continuous     | 16                          | 16  | 8   | 4   | 44    | 68.75%     |
|               | $2^2$ phase interval | 8                           | 4   | 4   | 4   | 20    | 31.25%     |
| 8             | Phase-continuous     | 16                          | 16  | 16  | 8   | 56    | 87.50%     |
|               | 2 phase interval     | 8                           | 8   | 8   | 8   | 32    | 50.00%     |

the computation nodes for the power-of-two phase interval output FFT is obtained. The output of the last stage, moving backward  $K$  stages, corresponds to  $2^K$  nodes. From stage  $K+1$  onwards, the number of nodes doubles with each preceding stage. Therefore, the total number of computation nodes for the power-of-two phase interval output FFT algorithm is given in Equation (14).

$$T_2 = K \times 2^K + 2^K \left( \sum_{i=0}^{L-K-1} 2^i \right) = (2^{L-K} - 1 + K) \times 2^K \quad (14)$$

In the FFT algorithm with power-of-two phase interval outputs, the last  $K$  stages consist of full butterfly operations, with  $2^{L-1}$  butterfly operations per stage. Each butterfly involves two complex additions and one complex multiplication, resulting in a total of  $K \cdot 2^K$  complex additions and  $K \cdot 2^{K-1}$  complex multiplications. Stages 1-th through  $K-1$ -th perform half-butterfly operations, where each computation node requires one complex addition and one complex multiplication, leading to  $2^L - 2^K$  complex additions and  $2^L - 2^K$  complex multiplications. In total, the algorithm performs  $K \cdot 2^K + 2^L - 2^K$  complex additions and  $K \cdot 2^{K-1} + 2^L - 2^K$  complex multiplications.

The total number of computation nodes for a conventional FFT is  $L \cdot 2^L$ . The ratio of computation nodes in the power-of-two phase interval output FFT relative to the conventional FFT is given in Equation (15).

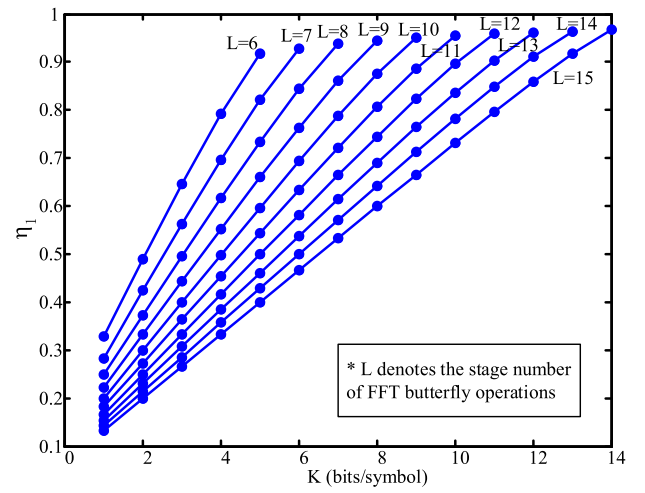
$$\eta_2 = \frac{T_2}{L \times 2^L} = \frac{2^{L-K} + K - 1}{L \times 2^{L-K}} \quad (15)$$

The proportion of computation in the power-of-two phase interval output FFT compared to the continuous phase partial-output FFT is given in Equation (16).

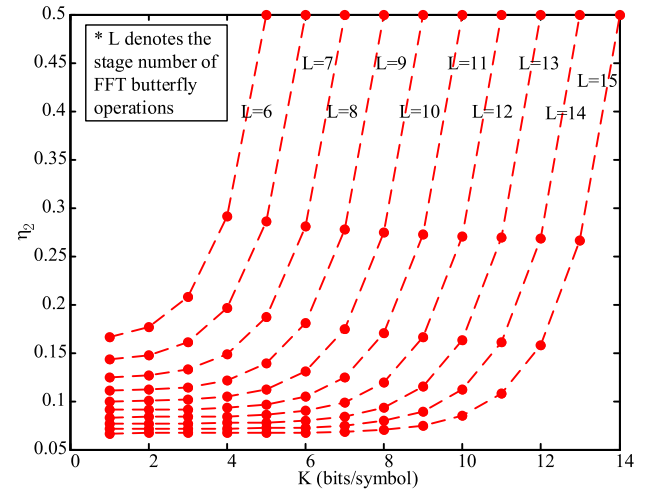
$$\eta_3 = \eta_1 - \eta_2 = \frac{1}{L} \left( K - 2^{K-L} - \frac{K-1}{2^{L-K}} \right) = \frac{1}{L} \left( K - \frac{K}{2^{L-K}} \right) \quad (16)$$

Based on Equations (8) and (15), the proportion of computation nodes required by the phase-continuous partial-output FFT and the power-of-two phase interval output FFT—relative to the total number of nodes in a conventional FFT—is calculated for different values of  $L$  and  $K$  (as shown in Figures 9 and 10). It is observed that for the power-of-two phase interval output FFT, when  $K$  is fixed, a larger input size  $L$  results in a smaller proportion of computation nodes relative to the total number in a full-output FFT.

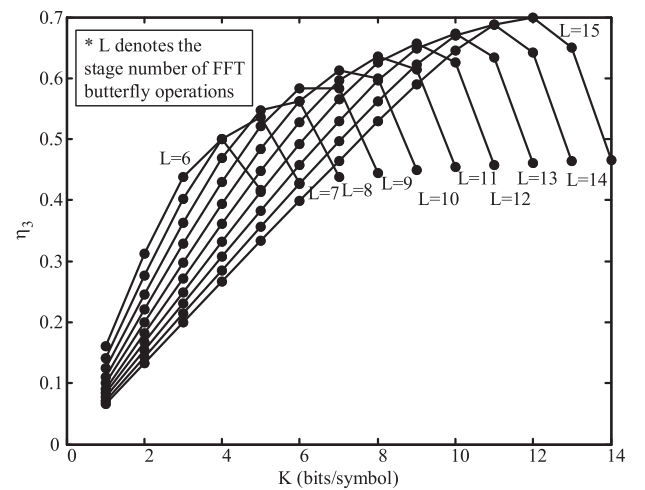
The difference in the proportion of computation nodes between the two optimisation algorithms can be calculated using Equation (14), as shown in Figure 11. It is observed that the computation of the power-of-two phase interval output FFT algorithm is smaller than that of the continuous phase partial-output FFT algorithm. Moreover, when  $K = L-3$ , the efficiency difference between the two reaches its maximum. For example, when  $L = 15$  and  $K = 12$ , the proportion of computation nodes in the total nodes for the partial-output FFT algorithm is 85.83%, whereas the proportion for the power-of-two phase interval output FFT algorithm is 15.83%. In this case, the latter saves more computation compared to the former.



**FIGURE 9** | The computation node proportions of phase-continuous partial-output FFT based on Equation (8).



**FIGURE 10** | The computation node proportions of phase-discontinuous power-of-two interval output FFT based on Equation (15).



**FIGURE 11** | Difference in computation node proportions between partial-output FFT and power-of-two interval output FFT based on Equation (16).

## 5 | Application in BDS PPP-B2b Signal Design

The PPP service of BDS-3 is broadcast via the PPP-B2b signal transmitted by GEO satellites. The carrier frequency of the PPP-B2b signal is 1207.14 MHz and comprises an I-component and a Q-component. Currently, only the I-component is being transmitted, using BPSK(10) modulation with a ranging code rate of 10.23 Mcps, a code length of 10,230 chips, a symbol rate of 1000 sps and a data rate of 500 bps, employing 1/2-rate 64-ary LDPC channel coding [4]. In the future, the data rate of the PPP-B2b signal is expected to be upgraded to 1000 bps or even 2000 bps [5] in order to support higher-precision and faster-converging PPP services at decimetre- and centimetre-level accuracy. The Q-component signal of the PPP-B2b signal employs CSK modulation with power-of-two phase interval mapping, as proposed in this paper. This method not only increases the data transmission rate but also mitigates intersymbol interference and decreases the computational complexity of CSK demodulation at the receiver.

### 5.1 | Generation of PPP-B2b Signal

The signal generation principle of the PPP-B2b Q-component employing CSK modulation with power-of-two phase interval mapping is illustrated in Figure 12. The Q-component of the PPP-B2b signal adopts a repeated code shift keying (CSK (6,2)) scheme to modulate the message data, with a ranging code rate of 10.23 Mcps, a code length of 10,230 chips, a symbol rate of 3000 sps and a data rate of 1500 bps. It employs 1/2-rate 64-ary LDPC channel coding. As a result, the combined data broadcast capacity of the I-component and Q-component of the PPP-B2b signal reaches 2000 bps.

According to Equation (9),  $P = \lceil \log_2(10230) \rceil - 6 = 7$ , and the phase interval is denoted as  $Z = 2^P = 128$  chips. Figure 13 depicts the mapping relationship between the modulation information and the phase of the spreading sequence. Figure 14 depicts the corresponding modulation timing diagram.

Considering a phase interval of 128 chips, it can be deduced that when the multipath delay is within the range of 1–127 chips (equivalent to 97.75 ns to 12.41  $\mu$ s), no modulation symbol is assigned to the corresponding delayed phases within this range. This effectively prevents intersymbol interference. Under identical configuration parameters, CSK signals with continuous phase mapping are unable to suppress ISI, whereas CSK signals

with equal-phase-interval mapping can eliminate ISI resulting from multipath delays ranging from 1 to 154 chips [15]. Although the interference mitigation range of the power-of-two phase interval mapping CSK signal is somewhat narrower than that of the equal-phase-interval scheme, it is adequate to satisfy the requirements of most practical scenarios.

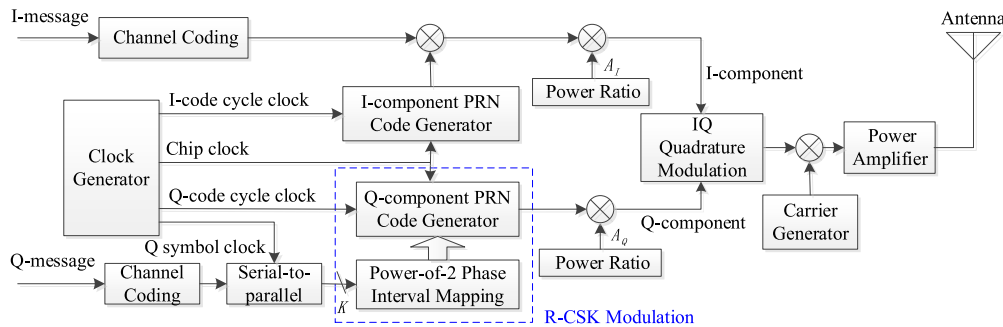
### 5.2 | Reception and Demodulation of PPP-B2b Signal

The demodulation of CSK-modulated PPP-B2b signals with power-of-two phase intervals is illustrated in Figure 15 [15]. The ‘Comb Filter’ component aggregates N sets of spreading sequence data within the same symbol into a single set of spreading sequence data. The ‘Frequency-Domain Matched Filtering’ component conducts frequency-domain matched filtering on the incoming spreading sequence data and the local spreading sequence data, subsequently outputting a sequence of correlation values. The ‘Maximum Value Search’ component executes a maximum-value determination on the input sequence of correlation values and yields the CSK modulation phase. The ‘Message Mapping’ component demodulates the data information of the Q-component signal in accordance with the mapping relationship between the modulation phase and the modulation data.

The PPP-B2b Q-component signal can be received using a frequency-domain demodulation method based on power-of-two phase interval output FFT, as illustrated in Figure 6. For

| Message<br>(Binary) | Message<br>(Decimal) | PRN code<br>phase |      |      |     |       |       |  |
|---------------------|----------------------|-------------------|------|------|-----|-------|-------|--|
| 000000              | 0                    | PRN(0)            | 0    | 1    | ... | 10228 | 10229 |  |
| 000001              | 1                    | PRN(128)          | 128  | 129  | ... | 126   | 127   |  |
| 000010              | 2                    | PRN(256)          | 256  | 257  | ... | 254   | 255   |  |
| ⋮                   | ⋮                    | ⋮                 |      |      |     |       |       |  |
| 111111              | 63                   | PRN(8064)         | 8064 | 8065 | ... | 8062  | 8063  |  |

**FIGURE 13** | Mapping relationship between modulation information and PRN phases of CSK-modulated PPP-B2b\_Q signal.

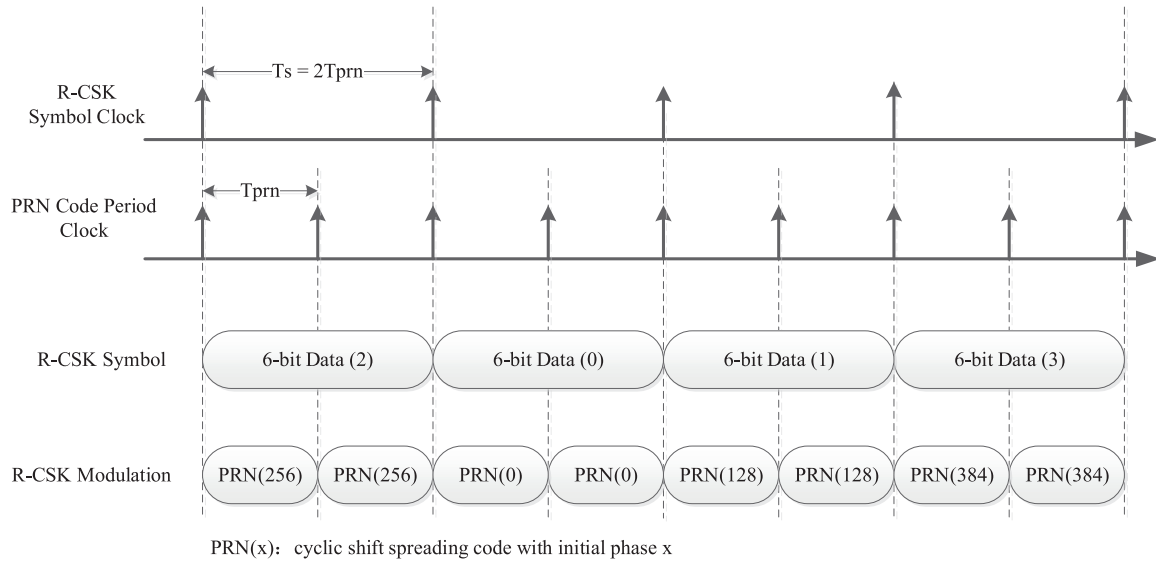


**FIGURE 12** | Block diagram of PPP-B2b signal generation based on power-of-two phase interval CSK modulation.

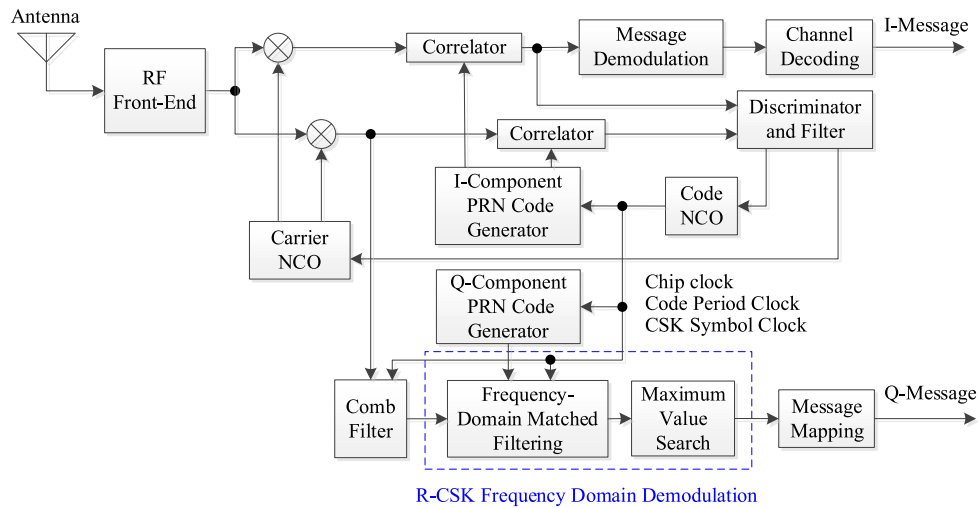


both the synchronised and carrier-stripped baseband signal sequence  $s(n)$  and the basic spreading sequence  $c_d(n)$ , zero-padding is applied to extend the sequence length from 10,230 to 16,384 ( $L = \lceil \log_2 10230 \rceil = 14$ ,  $N = 2^L = 16384$ ).

Considering the phase interval of 128 chips, the number of FFT output points for the power-of-two phase interval mapping is 128 ( $16,384/128 = 128$ ). From these, the first 64 points are the desired phase correlation values. Table 2 provides a comparative



**FIGURE 14** | CSK modulation timing diagram of B2b\_Q signal.



**FIGURE 15** | Block diagram of frequency-domain demodulation for CSK-modulated PPP-B2b signal with power-of-two phase intervals.

**TABLE 2** | Comparison of computation nodes for the three mapping methods.

| Mapping method              | Phase-continuous mapping | Equal phase interval mapping | Power-of-two phase interval mapping    |
|-----------------------------|--------------------------|------------------------------|----------------------------------------|
| Demodulation method         | Partial-output FFT       | Full-output FFT              | Power-of-two phase interval output FFT |
| FFT points                  | 16,384                   | 16,384                       | 16,384                                 |
| Butterfly stages            | 14                       | 14                           | 14                                     |
| Output points               | 64                       | 16,384                       | 128                                    |
| Computation nodes           | 114,624                  | 229,376                      | 17,152                                 |
| Phase interval              | 1                        | 155                          | 128                                    |
| Computation node proportion | 49.97%                   | 100%                         | 7.48%                                  |

analysis of the computational complexity among the power-of-two phase interval output FFT method, the conventional full-output FFT method and the phase-continuous partial-output FFT method. It is clear that the proposed algorithm requires only 7.48% of the computation nodes of the conventional FFT and 15% of those of the partial-output FFT. This translates to an approximate 85% reduction in demodulation computation compared to the phase-continuous partial-output FFT method.

## 6 | Conclusion

This paper addresses the issue of intersymbol interference in message demodulation due to multipath interference in CSK signals, as well as the exponential increase in reception complexity with the rise in the number of bits per symbol. A novel approach for the design and reception of CSK signals with a phase interval mapping of power-of-two chips is proposed, building upon phase-discontinuous equal-interval mapping. This approach significantly reduces demodulation complexity while mitigating intersymbol interference in CSK signal message demodulation. The method is applied to the design of the BDS PPP-B2b service signal. The CSK (6, 2) modulation with a phase interval mapping of 128 chips is employed for the Q-component signal, which has a spreading sequence code rate of 10.23 Mcps and a chip length of 10,230. This method can eliminate intersymbol interference caused by multipath delays ranging from 97.75 ns to 12.41  $\mu$ s. The signal is demodulated using the FFT with a phase interval output of 128 points. Compared to the phase-continuous mapping partial-output FFT demodulation method, it reduces demodulation computation by approximately 85%. This approach provides valuable insights for the design of satellite-based PPP signals for next-generation satellite navigation systems.

### Author Contributions

**Nengjie Yu:** conceptualization, methodology, writing – original draft, writing – review and editing. **Dun Wang:** conceptualization, formal analysis, supervision, writing – review and editing. **Xiaohui Ba:** funding acquisition, investigation, project administration, validation. **Minquan Lu:** formal analysis, resources, supervision. **Shaobo Shen:** data curation, validation, visualization.

### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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